

REASONING BASED DI – 10-9-2026

Directions for questions 1 to 5: Answer these questions on the basis of the information given below.

150 students of a school were selected to attend workshop in Music, Drawing and Acting. The students had the option of attending one or more workshops. It is known that the number of students who attended the music workshop was more than the sum of the number of students who attend the workshop on drawing and the number of students who attended the workshop on acting.

What is the maximum number of students who attended only the acting workshop?

- ☐ a) 49
- ☐ b) 50
- ☐ c) 74
- ☐ d) 75

Solution:

As the total number of students is 150 and $n(\text{music}) > n(\text{drawing}) + n(\text{acting})$, the maximum number of students who attended only the acting workshop would be 74 (while 76 attended the music workshop).
Choice (C)

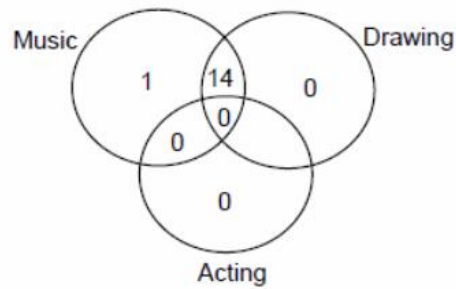
What is the maximum number of students who attended the drawing workshop?

- ☐ a) 74
- ☐ b) 75
- ☐ c) 100

☐ d) 149 (Correct Answer - ✓)

Solution:

The maximum number of students who attended the drawing workshop would be 149 as shown below



Choice (D)

What is the maximum number of students who attended all the three workshops?

☐ a) 74 (Correct Answer - ✓)

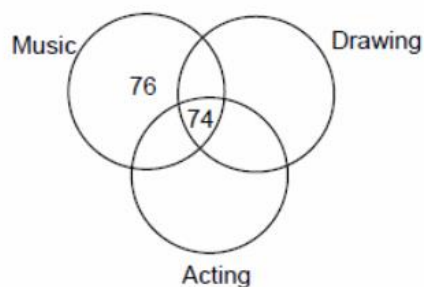
☐ b) 75

☐ c) 100

☐ d) 149

Solution:

The maximum number of students who attended all the three workshops would be 74. Any number greater than 74 would mean that the conditions $n(\text{music}) > n(\text{drawing}) + n(\text{acting})$ is not maintained.



Now $(76 + 74) > (74) + (74)$

Choice (A)

What is the minimum number of students who attended the music workshop?

- ☐ a) 0
- ☐ b) 51
- ☐ c) 75

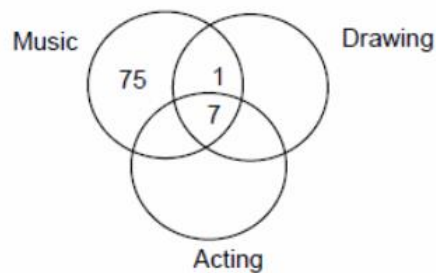
☐ d) 76 (Correct Answer - ✓)

Solution:

The minimum number of students who attended the music workshop would be 76 (as in the first question) Choice (D)

Solution:

The maximum number of people who attended the acting workshop would be 74 as follows



Here music is 150, drawing is 75 and acting is 74 and $150 > 75 + 74$ Choice (C)

Directions for questions 6 to 10: Answer these questions on the basis of the information given below.

The following table gives the number of people getting in and getting out of a bus at four stops:- P, Q, R and S. P is not the first stop and all people get down at S, the last stop. In each stop, the people first get down before people get in to the bus.

Stop	Getting in	Getting out
P	25	23
Q	27	25
R	22	34
S	--	

Directions for questions 6 to 10: Type in your answer in the input box provided below the question.

What is the minimum number of people who got down at stop S?

Solution:

All people who got in at P and Q could have got down at R.
 $\therefore$ at least 22 people (all those who got in at R) would have got down at S.
Ans : (22)

Directions for questions 6 to 10: Type in your answer in the input box provided below the question.

At least how many people were in the bus when it reached stop P?

Your Answer: Unattempted

Show Correct Answer

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PREVIOUS

NEXT

Solution:

As 52 ($25 + 27$) got in at P or Q and 59 ($25 + 34$) got down at Q or R, these extra seven people were surely in the bus when it reached Q. As 23 people got down at Q, at least $23 + 7 = 30$ people were in the bus when it reached P.
Ans : (30)

Directions for questions 6 to 10: Type in your answer in the input box provided below the question.

At least how many people who got in at P got down at stop S?

Your Answer: Unattempted

Show Correct Answer

Hide Solution

PREVIOUS

NEXT

Solution:

All people who got in at P would have got down at Q or R. $\therefore$ there need not be anybody, who got in at P and got down at S.
Ans : (0)

Directions for questions 6 to 10: Type in your answer in the input box provided below the question.

If there were 30 people in the bus when it reached stop P, what is the minimum number of people who got down at the next stop they got in?

Your Answer: Unattempted

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Solution:

Of the 30 people in the bus when E reached P, 23 got down at P. The remaining 7 would get down at Q along with $18(25 - 7)$ people who got in at P. At R, the remaining 10 people who got in at Q along with 24 people who got in at Q would get down at R. At S all of them would get down.
 $\therefore$ Minimum of $18 + 24 + 22 = 64$ people got down at the next stop they got in.
Ans : (64)

Directions for questions 6 to 10: Type in your answer in the input box provided below the question.

If all persons who got in at stop P got down at stop S, what is the minimum number of persons in the bus when it reached stop P?

Your Answer: Unattempted

Show Correct Answer

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PREVIOUS

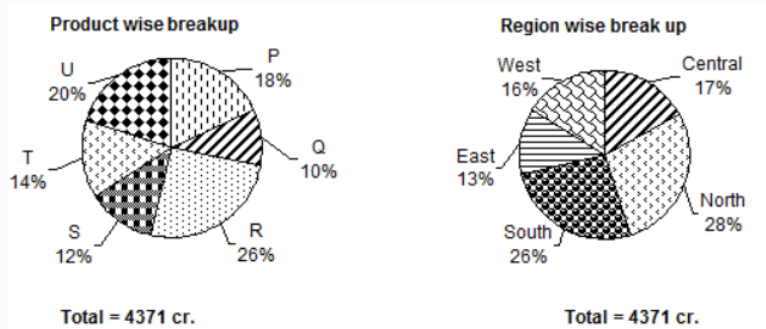
NEXT

Solution:

If all people who got in at P, got down at S, the bus should have had 23 (people who got down at P) + 25 (people who got down at Q) + $7(34 - 27) = 55$ people were in the bus when it reached P.
Ans : (55)

Directions for questions 11 to 15: Answer these questions on the basis of the information given below.

The following pie charts give the breakup of sales (by value) of all the six products of a company and its sales region wise



At most how many products were sold only in the north region?

- ☐ a) 1
- ☐ b) 2
- ☐ c) 3
- ☐ d) 4

Solution:

As the north region contributes 28% to the total sales, at most two products (P and Q or Q and T or Q and S) were sold only in the north region.

Choice (B)

At least how many of the six products were sold in the south region?

- ☐ a) 1
- ☐ b) 2
- ☐ c) 3
- ☐ d) 4

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N

Solution:

The entire sales of R could be in the south region in which case, it would be the only product sold in the south region.
Choice (A)

If products S, T and Q were not sold in the north and south regions, at least what percentage of product T was sold in the central region?

- ☐ a) 0
- ☐ b) 5
- ☐ c) 12
- ☐ d) None of these

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Solution:

S, T and Q was sold in east, west or the central regions. Product T could be entirely sold in the east or west region and need not be sold in the central region at all.
Choice (A)

If at least 10% of the sale of each product was from the central region, at most what percentage of any product was sold in the central region?

- ☐ a) 20%
- ☐ b) 50%
- ☐ c) 70%
- ☐ d) 80%

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Solution:

10% of total sales was from the central region. Now as Q has the lowest sales among all the products another 7%(17% - 10%) of Q can be sold in the central region. $\therefore$ 80% of Q (1% + 7%) could be, sold in the central region. Choice (D)

At most how many of the given products were sold in exactly one region?

- ☐ a) 3
- ☐ b) 4
- ☐ c) 5
- ☐ d) 6

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PREVIOUS

Solution:

P and Q could be from north alone, R from south alone, S from east alone and T from east alone. $\therefore$ at most five products could be sold in one region alone.

Choice (C)